1: Introduction to Technical Writing

CHAPTER OBJECTIVES

- Define technical communication.
- Distinguish technical writing from academic writing.
- Explain the features of technical writing style.
- Introduce the concepts of audience, culture, and ethics as they apply to technical writing.

What is Technical Writing?

You are probably wondering what this "technical writing thing" is. Someone may have even told you, "It's this course where they make you write about rocket science and brain surgery." Well, not really, as you will see in a moment. The field of technical communication is essential in a wide range of fields and occupations. It is a fully professional field with degree programs, certifications, and—yes!—even theory. It is a good field with a lot of growth and income potential, and an introductory technical-writing course for which this book has been developed is a good way to start if you are interested in a career in this field.

Technical writing is designed for users in technical fields such as engineering, chemistry, computer information software and systems, medical professions, aeronautics, robotics, automotive, etc.

Technical writing is designed to:

- inform by anticipating and answering audience questions;
- instruct the audience to perform a task or follow a procedure, and
- persuade the audience via explanations, analysis, and solutions.

Technical writing is an audience-centered means of communication that provides a reader with clear and easy access to information. In the business world, time equates to profit, and profit is the force behind all professional interactions. The technical writer and reader have a vis-à-vis relationship. The writer recognizes, respects, and addresses the importance of time in effective and efficient communication by providing documents written in specific formats, using unambiguous language to send clearly accessible information. The reader in turn thoroughly understands the information in order to give a thoughtful response.

CHARACTERISTICS OF TECHNICAL WRITING

Technical communication is an audience-centered means of communication that provides the reader with clear, accurate, and ethically represented information. According to the Society for Technical Communication, technical writing includes the following characteristics:

- “Communicating about technical or specialized topics, such as computer applications, medical procedures, or environmental regulations.”
- “Communicating by using technology, such as web pages, help files, or social media sites.”
- “Providing instructions about how to do something, regardless of how technical the task is or even if the technology is used to create or distribute that communication.”

The Meaning of "Technical"

Technical communication—or technical writing, as the course is often called—is not writing about a specific technical topic such as computers, but about any technical topic. The term "technical" refers to knowledge that is not widespread, that is more the territory of experts and specialists. Whatever your major is, you are developing an expertise—you are becoming a specialist in a particular technical area. And whenever you try to write or say anything about your field, you are engaged in technical communication.

Academic Writing Versus Technical Writing

Technical communication is distinct from the academic forms of writing you may be more familiar with. The academic writer’s purpose may be to write an essay, a story, a research paper, etc. Such assignments are often designed so that students can “write to learn” and show mastery of information covered in class. Additionally, in academic writing context, student-writers join a conversation that is occurring on a particular topic.

Technical communication, however, is intended for another purpose. These documents convey information to audiences who may or may not have prior knowledge of the material discussed. Technical communicators must, therefore, determine the needs of their audience and design documents that convey information in an accessible and appropriate manner. Depending on the context of communication, it might also be necessary to convey information in a concise and efficient manner, succinctly presenting points and cutting extraneous or potentially distracting material.

Workplace Writing

However, the focus for technical writing courses is not necessarily a career as a technical writer but an introduction to the kinds of writing skills you need in practically any technically-oriented professional job. No matter what sort of professional work you do, you're likely to do lots of writing—and much of it technical in nature. The more you know about some basic technical-writing skills, which are covered in this guide and in technical-writing courses, the better job of writing you're likely to do. And that will be good for the projects you work on, for the organizations you work in, and—most of all—good for you and your career.

STRATEGIES TO ACHIEVE TECHNICAL WRITING TASKS

Professional technical writers rely on these strategies to ensure the technical accuracy of their work:

- Study of books, articles, reports, websites related to the product.
- Product specifications: what the product is supposed to do, how it is designed.
- Interviews with subject matter experts: the product specialists, developers, engineers.
- Product meetings during the development cycle.
- Live demonstrations of the product.
- Familiarization with similar, competing products.
- Experimenting with working models of the product.
- Most importantly, subject matter experts' review of technical writers' work for technical accuracy and completeness.

Considerations of Technical Documents

There are key components of what makes a document strong. Therefore, writers keep these items in mind while constructing technical documents.

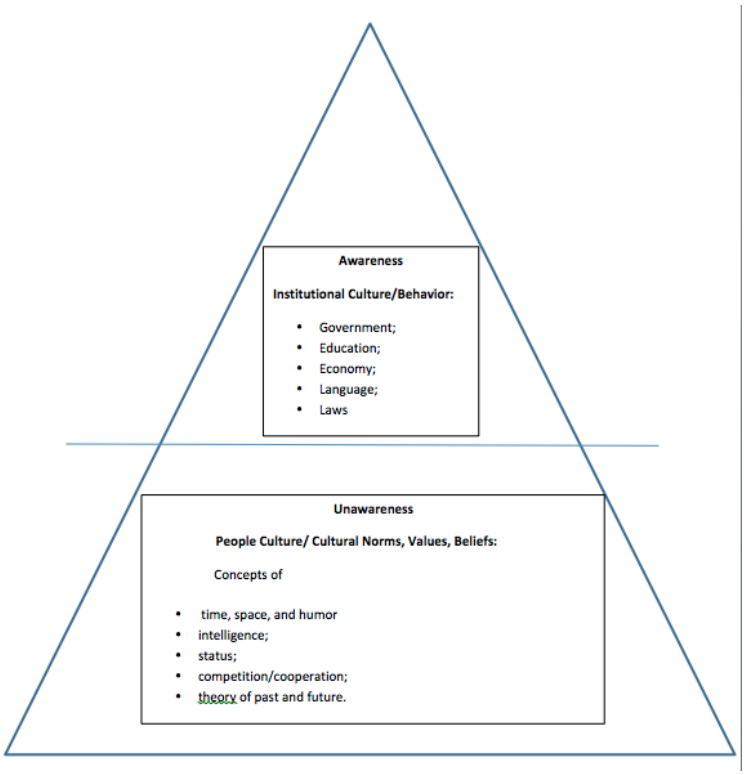
The Importance of Audience

Another key part of the definition of technical communication is the receiver of the information—the audience. Technical communication is the delivery of technical information to readers (or listeners or viewers) in a manner that is adapted to their needs, level of understanding, and background. In fact, this audience element is so important that it is one of the cornerstones of this course: you are challenged to write about highly technical subjects but in a way that a beginner—a nonspecialist—could understand. This ability to "translate" technical information to non-specialists is a key skill to any technical communicator. In a world of rapid technological development, people are constantly falling behind and becoming technological illiterates. Technology companies are constantly struggling to find effective ways to help customers or potential customers understand the advantages or the operation of their new products.

Cultural Communication

Technical writers need to be aware of the differences between the behavior and the norms, beliefs and values of specific cultural. According to Edward T. Hall and Mildred Reed Hall, In Understanding Cultural Differences, each culture operates according to its own rules (1990, pp. 3-4). Hall and Hall add that problems occur when members of one culture apply the rules to another culture (1990, pp. 3-4). To communicate effectively with other cultures, the technical writer needs to not only be aware of rules governing behaviors that can be observed but also of the not-so-obvious rules that govern the norms, beliefs, and values of the people of a culture. The invisible rules of a culture dramatically impact the acceptance of ideas, plans, and strategies. The Cultural Iceberg illustrates patterns of world communication, showing indicators of Institutional Culture (the obvious behavior of a culture), which can be clearly seen as the tip of the iceberg, and People Culture (the norms, beliefs and values of a culture), which cannot be seen and which are the barriers to successful communication.

Figure 1 The Cultural Iceberg



Ethics

Technical writers have a responsibility to their readers and to their employers to follow ethics when writing reports. Technical writers must use words that demonstrate valid appeals to reason, avoiding emotional words and phrases that appeal to basic emotion instead of justifiable reasoning. In addition, technical writers must use valid references to support ideas and strategies, avoiding referencing non experts to sway readers' support. Also, technical writers must use accurate numbers to report data, avoiding charts and tables that skew data. Using any type of fallacies in technical writing is unethical and could result in dire consequences.

Not only do technical writers have a responsibility to report accurate information, but they also have a responsibility to credit accurate sources of information. At no time is it acceptable to rearrange information in order to attempt to indicate that the writer is the source of someone else's idea or to indicate that the writer read a report that included information he/she cited, when the primary source of the information was cited in another report. All sources must be referenced accurately in the text and cited on a reference page.

Overview of the Technical Writing Style

Technical writing takes complicated ideas and organizes and explains those ideas in easy-to-understand language. The reader of technical documents does not read to engage in a discussion or be entertained. The audience of technical documents wants information. In short, technical writing provides information and seeks to solve problems and help the reader learn.

While some technical writing is composed for experts within the field, most technical writing is composed for the ordinary user - the consumer. While technical writing can organize information so it persuades the user, it must always present complete and accurate information. Technical writing does not use emotionally charged language, redundant adjectives, colloquialisms, or words or phrases that are open to interpretation. As the Society of Technical Writing explains, technical communicators employ a user-centered approach to provide "the right information, in the right way, at the right time to make someone's life easier and more productive."

Features of Technical Writing

Technical documents must be reader-centered. The information is explained and presented in a style that is easy to navigate and understand. Technical documents value the reader's time by using the following features:

- Accessible – Think about the users of your document. You understand the material and created the document based on that knowledge and understanding, but for your audience, this is new information. So the information must be explained and presented in a style that is easy to understand and follow.
- Collaborative - technical documents must consider multiple perspectives; therefore, they are frequently composed by a team of writers.
- Concise –To write concisely does not mean to use fewer words. Instead, it means to use *all* the words you need but *only* the words you need.
- Efficient page design: Use headings, numbered or bulleted lists, tables, easy-to-read fonts, white space, and other elements to help the reader navigate through the material.
- Logical organization: use chronological order and emphasize important information.
- Meaningful content: include all of the information needed but none of the information that is not needed.
- Supplemental material: Abstracts, footnotes, glossary, appendix, definitions, etc., provides readers with additional information when needed.
- Visual elements: Use charts, graphs, or images to clarify written concepts or relationships.

Formatting and Language

Formatting and appropriate language are the basic design elements of all technical documents. A format that shows a hierarchical structure and a coordinate structure of information leads the reader through the text. Readers should be able to identify a writer's organizational pattern very quickly when reading a technical document. This sometimes refers to a document being "reader-friendly." In addition, using appropriate language is significant in providing the reader with a thorough understanding of the purpose of the document, how the document relates to the reader's needs, and what action is expected of the reader.

A document may also have one audience (the primary reader) or multiple audiences (the secondary readers). A primary reader is the person who ordered the report to be written or the person for whom the report is intended. These readers will usually read the entire report. Secondary readers are those readers who will read only the sections of the report that relate to them, their jobs, their departments, responsibilities, etc. For example, if a report was sent that detailed funding for different departments, a piping superintendent may only want to read the section that relates to piping. This is where format, the use of headings, is significant in allowing the reader easy access to information. When the piping superintendent can scan through the document and clearly find the heading that identifies his department saves time.

The following table provides general specifications for many types of technical writing documents:

Document Formatting Rules

MARGINS	**Use 1″ margins** on all sides (use 2″ when binding)
	Justify your left margin only; don't fully justify your paragraphs, as this can result in odd spacing
FONTS	**Headings**: *Sans serif*, such as Arial or Calibri
	Body text: *Serif* font, such as Times New Roman or Cambria
FONT SIZE	**Headings**: 12-20 point *sans serif* font
	Body text: 11-12
SPACING	**Single-spacing** is used for most letters, memos, and emails; 1.5 or double spacing to allow for comments.
LENGTH	**Paragraphs** tend to be no longer than 10 lines
	Sentences are usually 15-20 words

Table 3.1. Document Formatting

Readability in Publications

The way a text looks matters to a reader, so it should matter to a writer. Letters, reports, and websites are more than just words on a page or a screen. How ideas are arranged and delivered in physical form, whether electronically or on paper, can make reading seem intimidating, confusing, or downright unfriendly, even if the content itself is perfect. Your text is like a room for your ideas. Sometimes you want readers to get in and get out quickly, but often, you want them to sit down and make themselves comfortable, put their feet up and stay awhile. Whatever the case, you should be in control of the reader's experience.

To make a document more reader-friendly, many technical writers rely on visuals to achieve this goal. See Chapter 15, "Visuals in Technical Documents" for detailed information about using visuals.

Ways effective design enhances a document

Effective design makes a document accessible

- Good design helps readers understand your information.
- Good page design helps readers locate information quickly.
- Good design helps readers notice highly important content.

Effective design affects readers' attitudes, thereby increasing a communication's persuasiveness.

- Good design encourages readers to feel good about the communication itself.
- Good design encourages readers to feel good about the communication's subject matter.

Because page design can have such a significant impact on your communication's usability and persuasiveness, you should approach design in the same reader-centered manner that you use when drafting text and graphics. Think continuously about your readers, including who they are, what they want from your communication and the context in which they will be reading.

It helps to think about the building blocks of a page design in the way that professional graphic designers do. When they look at a page, they see six basic elements:

- **Text**: Paragraphs and sentences.
- **Headings and titles**: Labels for sections of your communication.
- **Graphics**: Drawings, tables, photographs, and so on — including their captions.
- **White space**: Blank areas.
- **Headers and footers**: The items, such as page numbers, that occur at the top or bottom of each page in a multi-page document.
- **Physical features**: These include paper, which may take many shapes and sizes, and bindings, which come in many forms.

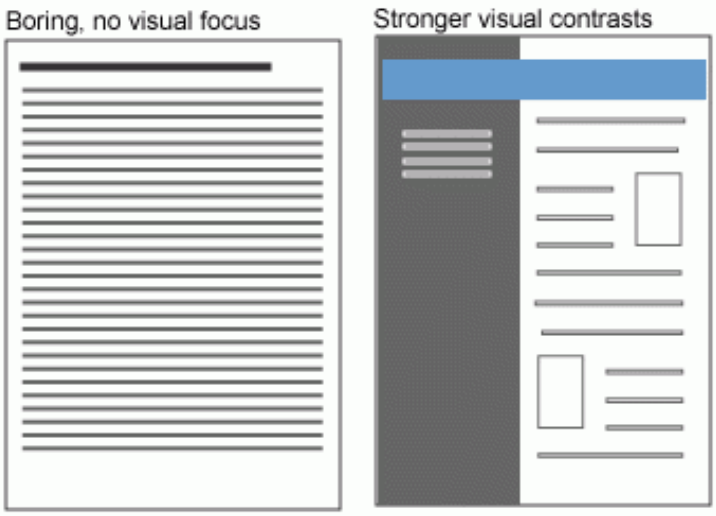


Figure 3.1. Overview of design elements. *Notice how your eye is drawn to the blue header and the boxed elements. In these spaces, you can highlight the important parts of your message:*

Conclusion

Technical writing is designed to inform, instruct, or persuade an audience. It is informative and concise. Technical writers often collaborate with others in their organizations to develop documents that are formatted and designed to inform their audience in accessible ways. All forms of technical writing -- memos, letters, proposals, reports, and so on -- have different conventions for format and design. Technical writing differs from the academic writing to which many students are accustomed. As you continue exploring concepts of technical communication in this course, remember these introductory concepts to technical writing and work to apply them to the documents you create.

GENERAL TIPS ABOUT TECHNICAL WRITING

- Remember that technical writing is not just about computers or engineering. The term should be considered more broadly than that.
- Audience is crucial. Before creating any technical document, be certain you understand your audience's identity and needs.
- Format your technical documents differently than you might format an academic writing assignment.
- Consider cultural and ethical concerns and complications as you develop your documents.

References

Hall, E.T. & Hall, M.R. (1990). *Understanding Cultural Differences.* Intercultural Press.

Society for Technical Communication. (2021). *Defining Technical Communication.* https://www.stc.org/about-stc/defini...communication/.

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